

无穷级数-知识总结

- 1.级数的性质
- 1.倍数不变敛散性:设 $c \neq 0$ 的常数, 则 $\sum_{n=1}^{\infty} u_n$ 与 $\sum_{n=1}^{\infty} cu_n$ 有相同敛散性

2.级数和的敛散性 $\left\{\begin{array}{l} \text{若}\sum_{n=1}^{\infty} u_n \text{收敛}, \sum_{n=1}^{\infty} v_n \text{发散, 则}\sum_{n=1}^{\infty} (u_n \pm v_n) \text{发散.} \\ \text{若}\sum_{n=1}^{\infty} u_n \sum_{n=1}^{\infty} v_n \text{均发散, 则}\sum_{n=1}^{\infty} (u_n \pm v_n) \text{敛散性不定.} \\ \text{若}\sum_{n=1}^{\infty} u_n = s, \sum_{n=1}^{\infty} v_n = \sigma, \text{则}\sum_{n=1}^{\infty} (u_n \pm v_n) = s \pm \sigma. \end{array}\right.$

3.加括号 $\left\{\begin{array}{l} \text{若}\sum_{n=1}^{\infty} u_n \text{收敛, 则}\sum_{n=1}^{\infty} (u_n + u_{n+1}) \text{收敛} \\ \text{若}\sum_{n=1}^{\infty} (u_n + u_{n+1}) \text{收敛, 则}\sum_{n=1}^{\infty} u_n \text{不一定收敛} \end{array}\right.$

- 4.绝对与条件收敛 $\left\{\begin{array}{l} \sum_{n=1}^{\infty} |u_n| \text{收敛, 则}\sum_{n=1}^{\infty} u_n \text{必收敛} \\ \sum_{n=1}^{\infty} u_n \text{条件收敛, 则}\sum_{n=1}^{\infty} \frac{u_n + |u_n|}{2}, \sum_{n=1}^{\infty} \frac{u_n - |u_n|}{2} \text{均发散} \\ \text{若}\sum_{n=1}^{\infty} u_n \text{绝对收敛, } \sum_{n=1}^{\infty} v_n \text{条件收敛, 则}\sum_{n=1}^{\infty} (u_n + v_n) \text{条件收敛} \end{array}\right.$

- 比较审敛法: 设 $0 \leq u_n \leq v_n$ $\left\{\begin{array}{l} \text{若}\sum_{n=1}^{\infty} v_n \text{收敛, 则}\sum_{n=1}^{\infty} u_n \text{收敛} \\ \text{若}\sum_{n=1}^{\infty} u_n \text{发散, 则}\sum_{n=1}^{\infty} v_n \text{发散} \end{array}\right.$
- 极限形式: $0 \leq u_n, \lim_{n \rightarrow \infty} \frac{u_n}{v_n} = A$ $\left\{\begin{array}{l} \text{若} 0 \leq A < +\infty, \text{且}\sum_{n=1}^{\infty} v_n \text{收敛, 则}\sum_{n=1}^{\infty} u_n \text{收敛} \\ \text{若} 0 < A \leq +\infty, \text{且}\sum_{n=1}^{\infty} v_n \text{发散, 则}\sum_{n=1}^{\infty} u_n \text{发散} \end{array}\right.$

- 2.正项审敛法
- 比值审敛法: 设 $0 \leq u_n$, 若 $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \rho$ $\left\{\begin{array}{l} \rho > 1 \text{时}, \sum_{n=1}^{\infty} u_n \text{发散} \\ \rho = 1 \text{时}, \text{方法失效} \\ \rho < 1 \text{时}, \sum_{n=1}^{\infty} u_n \text{收敛} \end{array}\right.$

根值审敛法: 设 $0 \leq u_n$, 若 $\lim_{n \rightarrow \infty} \sqrt[n]{u_n} = \rho$ $\left\{\begin{array}{l} \rho > 1 \text{时}, \sum_{n=1}^{\infty} u_n \text{发散} \\ \rho = 1 \text{时}, \text{方法失效} \\ \rho < 1 \text{时}, \sum_{n=1}^{\infty} u_n \text{收敛} \end{array}\right.$

- 3.常用参比级数
- 等比级数 $\sum_{n=1}^{\infty} ar^{n-1}$ $\left\{\begin{array}{l} \text{收敛, } |r| < 1 \\ \text{发散, } |r| \geq 1 \end{array}\right.$

p 级数 $\sum_{n=1}^{\infty} \frac{1}{n^p}$ $\left\{\begin{array}{l} \text{收敛, } p > 1 \text{时} \\ \text{发散, } p \leq 1 \text{时} \end{array}\right.$

交错版 p 级数 $\sum_{n=1}^{\infty} (-1)^n \frac{1}{n^p}$ $\left\{\begin{array}{l} p > 1, \text{绝对收敛} \\ 0 < p \leq 1, \text{条件收敛} \\ p \leq 0, \text{发散} \end{array}\right.$

- 4.非正项
- 交错级数: 莱布尼茨审敛法: $\left\{\begin{array}{l} \text{若}\sum_{n=1}^{\infty} (-1)^{n-1} u_n, (u_n > 0) \text{满足条件: (1)} u_n \geq u_{n+1}, (n = 1, 2, \dots): (2) \lim_{n \rightarrow \infty} u_n = 0 \\ \text{则交错级数}\sum_{n=1}^{\infty} (-1)^{n-1} u_n \text{收敛} \end{array}\right.$

任意项级数: $\left\{\begin{array}{l} \text{通项加绝对值后收敛} \Rightarrow \text{原级数绝对收敛 (此时原级数一定收敛)} \\ \text{通项加绝对值后发散, 但原级数收敛} \Rightarrow \text{原级数条件收敛} \\ \text{通项加绝对值后发散, 且原级数发散} \Rightarrow \text{原级数一定发散} \end{array}\right.$

- 5.级数证明题主要方法
- 1.比较审敛法
 题干中存在参比级数, 或和 p 级数/等比级数相关

2.级数和极限存在
 形如 $\sum_{i=1}^n (a_n - a_{n+1})$ 的级数, 通常用和极限存在证明其收敛

- 6.收敛半径
- 阿贝尔定理 $\left\{\begin{array}{l} \text{幂级数}\sum_{n=0}^{\infty} a_n x^n \text{在} x_0 \neq 0 \text{处收敛, 则其在满足 } |x| < |x_0| \text{ 的所有处绝对收敛} \\ \text{幂级数}\sum_{n=0}^{\infty} a_n x^n \text{在} x_1 \neq 0 \text{处发散, 则其在满足 } |x| > |x_1| \text{ 的所有处发散} \end{array}\right.$

收敛半径求法 $\left\{\begin{array}{l} \text{幂级数}\sum_{n=0}^{\infty} a_n x^n \text{收敛半径: } \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \rho \Rightarrow R = \frac{1}{\rho}. \\ \text{幂级数}\sum_{n=0}^{\infty} u_n(x) \text{的收敛区间: } \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \rho(x) < 1 \text{ 或 } \lim_{n \rightarrow \infty} \sqrt[n]{|u_n(x)|} = \rho(x) < 1, \text{ 解出的} x \text{ 范围} \\ \text{幂级数}\sum_{n=0}^{\infty} u_n(x) \text{的收敛域: 判断收敛区间端点的收敛性} \end{array}\right.$

- 7.幂级数展开
- 直接法: $f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \dots$ $\left\{\begin{array}{l} \frac{1}{1-x} = 1 + x + x^2 + \dots + x^n + \dots = \sum_{n=0}^{\infty} x^n, (-1, 1) \\ e^x = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \dots = \sum_{n=0}^{\infty} \frac{x^n}{n!}, (-\infty, +\infty) \\ \sin x = x - \frac{x^3}{3!} + \dots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \dots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}, (-\infty, +\infty) \\ \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + (-1)^n \frac{x^{2n}}{(2n)!} + \dots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}, (-\infty, +\infty) \\ \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots + (-1)^n \frac{x^{n+1}}{n+1} + \dots = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}, (-1, 1) \\ (1+x)^a = 1 + ax + \frac{a(a-1)}{2!} x^2 + \dots + \frac{a(a-1) \cdots (a-n+1)}{n!} x^n \cdots, (-1, 1) \end{array}\right.$

间接法:

- 求解思路
- ① n 在分子: 先积分再求导(如 $\sum_{n=1}^{\infty} nx^n$ 变形为 $x \sum_{n=1}^{\infty} nx^{n-1}$, 先积分再求导)

② n 在分母: 先求导再积分(如 $\sum_{n=1}^{\infty} \frac{x^n}{n}$, 先求导再积分)

③出现递推关系, 首选构建微分方程求级数之和

8.级数求和:

基本模型

- 最重要模型: $\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \dots + x^n + \dots = \frac{1}{1-x}$
- 次重要模型: $\sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1} = x - \frac{x^2}{2} + \frac{x^3}{3} - \dots + (-1)^n \frac{x^{n+1}}{n+1} + \dots = \ln(1+x)$
- 带阶乘模型: $\left\{\begin{array}{l} \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \dots = e^x \\ \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \dots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \dots = \sin x \\ \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + (-1)^n \frac{x^{2n}}{(2n)!} + \dots = \cos x \\ 1 + \sum_{n=1}^{\infty} \frac{a(a-1) \cdots (a-n+1)}{n!} x^n = 1 + ax + \dots + \frac{a(a-1) \cdots (a-n+1)}{n!} x^n \cdots = (1+x)^a \end{array}\right.$

9.傅里叶 (仅数一)

- 1.狄利克雷定理

- 设周期为 2π 的函数 $f(x)$ 在 $[-\pi, \pi]$ 上满足条件:
(1)除有限个第一类间断点外都连续;(2)只有有限个单调区间, 则
则 $f(x)$ 的以 2π 为周期的傅立叶级数在 $[-\pi, \pi]$ 上收敛,
且 $\frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) = \frac{1}{2}[f(x^-) + f(x^+)]$

- 2.傅里叶展开

- $[-l, l]$ 上展开为傅里叶级数 $\left\{\begin{array}{l} a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi}{l} x dx, (n = 0, 1, 2, \dots) \\ b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi}{l} x dx, (n = 1, 2, \dots) \\ f(x) \sim \frac{1}{2} a_0 + \sum_{n=1}^{\infty} (a_n \cos \frac{n\pi}{l} x + b_n \sin \frac{n\pi}{l} x) \end{array}\right.$
- $[0, l]$ 上展开为正弦或余弦级数 $\left\{\begin{array}{l} \text{正弦系数} \begin{cases} a_n = 0 (n = 0, 1, 2, \dots) \\ b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx (n = 1, 2, \dots) \end{cases} \\ \text{余弦系数} \begin{cases} a_n = \frac{2}{l} \int_0^l f(x) \cos \frac{n\pi x}{l} dx (n = 0, 1, 2, \dots) \\ b_n = 0 (n = 1, 2, \dots) \end{cases} \end{array}\right.$
- 表达式: $f(x) \sim \frac{1}{2} a_0 + \sum_{n=1}^{\infty} (a_n \cos \frac{n\pi}{l} x + b_n \sin \frac{n\pi}{l} x)$